

LL(1) Grammars

- First 'L' stands for scanning input from left to right.
- Second 'L' for producing left most derivation.
- '1' for using one input symbol of lookahead at each step

1. Consider the grammar given below and perform LL(1) parsing.

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow (E) / id$$

Step 1: Eliminating left recursion

$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow +TE' / \epsilon \\ T &\rightarrow FT' \\ T' &\rightarrow *FT' / \epsilon \\ F &\rightarrow (E) / id \end{aligned}$$

$A \rightarrow \alpha A / \beta$
$A \rightarrow \beta A'$
$A' \rightarrow \alpha A' / \epsilon$

Step 2: Left factoring

→ Check whether productions having common symbol from left end.

Step 3: first and follow functions

$$\text{FIRST}(E) = \text{FIRST}(T) = \text{FIRST}(F)$$

$$= \{ (, id \}$$

$$\text{FIRST}(E') = \{ +, \epsilon \}$$

$$\text{FIRST}(T') = \{ *, \epsilon \}$$

$$\text{FOLLOW}(E) = \{ \$,) \}$$

$$\text{FOLLOW}(E') = \{ \$,) \} \rightarrow \left(\begin{array}{l} \text{Suppose follow of } E' \text{ does} \\ \text{not have symbol, then} \end{array} \right.$$

$$\text{FOLLOW}(T) = \{ +, \$,) \}$$

$$\text{FOLLOW}(T') = \{ +, \$,) \}$$

$$\text{FOLLOW}(F) = \{ *, +, \$,) \}$$

as per the rule 3, follow(E) can be taken as follow(E')

* (According rule 2, if right side does not have terminals we have to take first of E' as follow(T))

* If ϵ is there, we have to take follow(E)

Step 4: Predictive Parsing Table:

	+	*	(	)	id	\$
E			$E \rightarrow T E'$		$E \rightarrow T E'$	
E'	$E' \rightarrow + T E'$			$E' \rightarrow \epsilon$		$E' \rightarrow \epsilon$
T			$T \rightarrow F T'$		$T \rightarrow F T'$	
T'	$T' \rightarrow \epsilon$	$T' \rightarrow * F T'$		$T' \rightarrow \epsilon$		$T' \rightarrow \epsilon$
F			$F \rightarrow (E)$		$F \rightarrow id$	

→ If E is found in first, check follow position of E' non-terminal

Steps: Parsing the input string

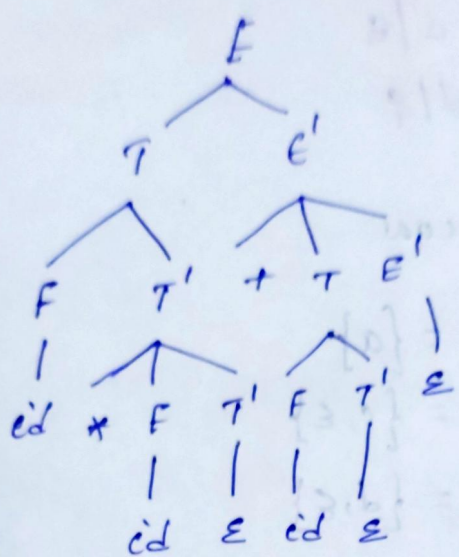
$$w = id * id + id$$

In stack o/p
should be
written in reverse.

Stack	Input	Output
\$ E	<u>id</u> * id + id \$	$E \rightarrow TE'$
\$ E' T	<u>id</u> * id + id \$	$T \rightarrow FT'$
\$ E' T' F	<u>id</u> * id + id \$	$F \rightarrow id$
\$ E' T' <u>id</u>	id * id + id \$	
\$ E' T'	* <u>id</u> + id \$	$T' \rightarrow * FT'$
\$ E' T' F *	* <u>id</u> + id \$	
\$ E' T' F	<u>id</u> + id \$	$F \rightarrow id$
\$ E' T' <u>id</u>	<u>id</u> + id \$	
\$ E' T'	+ <u>id</u> \$	$T' \rightarrow \epsilon$
\$ E'	+ <u>id</u> \$	$E' \rightarrow + TE'$
\$ E' T *	* <u>id</u> \$	
\$ E' T	<u>id</u> \$	$T \rightarrow FT'$
\$ E' T' F	<u>id</u> \$	$F \rightarrow id$
\$ E' T' <u>id</u>	id \$	$T' \rightarrow \epsilon$
\$ E' T'	\$	$E' \rightarrow \epsilon$

→ Bottom of stack contains \$ symbol. Initially bp of stack holds the start symbol of grammar

→ With the generated output draw the parse tree.



- (2) FIRST
- (A) FOLLOW
- (A) FOLLOW
- (3) FOLLOW
- (A) FOLLOW
- (A) FOLLOW

	a	b	c	d	

Handwritten notes and calculations at the bottom of the page, including various mathematical expressions and symbols.

3. Construct the parsing tree and check whether the grammar is LL(1) grammar or not.

$$S \rightarrow iELSS / iELSSa$$

$$E \rightarrow c$$

→ No left recursion

→ Left factoring

$$S \rightarrow \underline{iELS} / \underline{iELSS}a \quad (\text{Having common symbols from the left end})$$

$$\begin{array}{l} S \rightarrow iELSS' / a \\ S' \rightarrow \epsilon / eS \\ E \rightarrow c \end{array}$$

→ FIRST and FOLLOW

$$\text{FIRST}(S) = \{i, a\}$$

$$\text{FIRST}(S') = \{\epsilon, e\}$$

$$\text{FIRST}(E) = \{c\}$$

$$\begin{array}{l} \text{FOLLOW}(S) = \{\$, e, \text{FOLLOW}(S')\} \\ \quad = \{\$, e\} \\ \text{FOLLOW}(S') = \{\$, e\} \\ \text{FOLLOW}(E) = \{c\} \end{array}$$

→ Predictive Parsing Table

	i	t	a	e	c	$\$$
S	$S \rightarrow iELSS'$		$S \rightarrow a$			
S'				$S' \rightarrow \epsilon$ $S' \rightarrow eS$		$S' \rightarrow \epsilon$
E					$E \rightarrow c$	

$M[S', e] \Rightarrow a$ which is not LL(1) grammar

FIRST function

1. for a production rule $X \rightarrow \epsilon$,

$$\text{FIRST}(X) = \{\epsilon\}$$

2. for any terminal symbol 'a'

$$\text{FIRST}(a) = \{a\}$$

3. for a production rule $X \rightarrow Y_1 Y_2 Y_3$

* Calculating $\text{FIRST}(X)$

* If $\epsilon \notin \text{FIRST}(Y_1)$, then $\text{FIRST}(X) = \text{FIRST}(Y_1)$

* If $\epsilon \in \text{FIRST}(Y_1)$, then

$$\text{FIRST}(X) = \{\text{FIRST}(Y_1) - \epsilon\} \cup \text{FIRST}(Y_2 Y_3)$$

* Calculating $\text{FIRST}(Y_2 Y_3)$

→ If $\epsilon \notin \text{FIRST}(Y_2)$, then $\text{FIRST}(Y_2 Y_3) = \text{FIRST}(Y_2)$

→ If $\epsilon \in \text{FIRST}(Y_2)$, then

$$\text{FIRST}(Y_2 Y_3) = \{\text{FIRST}(Y_2) - \epsilon\} \cup \text{FIRST}(Y_3)$$

Follow function

1. for the start symbol S , place $\$$ in $\text{follow}(S)$
2. for any production rule $A \rightarrow \alpha B$
 $\text{follow}(B) = \text{follow}(A)$

3. for any production rule $A \rightarrow \alpha B \beta$

* If $\epsilon \notin \text{FIRST}(\beta)$, then $\text{follow}(B) = \text{FIRST}(\beta)$

* If $\epsilon \in \text{FIRST}(\beta)$, then

$$\text{follow}(B) = \{\text{FIRST}(\beta) - \epsilon\} \cup \text{follow}(A)$$

FIRST and FOLLOW

$$\begin{array}{l}
 S \rightarrow ABCD \\
 A \rightarrow a/\epsilon \\
 B \rightarrow b/\epsilon \\
 C \rightarrow AB/c \\
 D \rightarrow d
 \end{array}
 \left. \vphantom{\begin{array}{l} S \rightarrow ABCD \\ A \rightarrow a/\epsilon \\ B \rightarrow b/\epsilon \\ C \rightarrow AB/c \\ D \rightarrow d \end{array}} \right\} \begin{array}{l} \text{FIRST} \\ \text{FIRST}(S) = \{ \epsilon \} \\ \text{FIRST}(D) = \{ d \} \\ \text{FIRST}(B) = \{ b, \epsilon \} \\ \text{FIRST}(A) = \{ a, \epsilon \} \end{array}$$

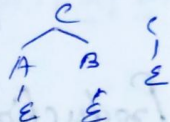
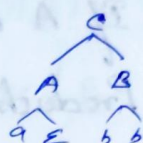
$$\text{FIRST}(C) = \text{FIRST}(AB) + \text{FIRST}(C)$$

$$= \{ a, \epsilon \}$$

$$\downarrow \text{FIRST}(B)$$

$$= \{ a, b, \epsilon \} + \{ c \}$$

$$= \{ a, b, c, \epsilon \}$$



$$\text{FIRST}(S) = \text{FIRST}(ABCD)$$

$$= \text{FIRST}\{ a, \epsilon \}$$

$$\downarrow \text{FIRST}(BCD)$$

$$= \{ a, b, \epsilon \}$$

$$\downarrow \text{FIRST}(CD)$$

$$= \{ a, b, c, \epsilon \}$$

$$\downarrow \text{FIRST}(D)$$

$$= \{ a, b, c, d \}$$

FIRST

$$\textcircled{2} E \rightarrow TE'$$

$$E' \rightarrow +TE' / \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' / \epsilon$$

$$F \rightarrow (E) / id$$

$$\begin{aligned} FIRST(E) &= FIRST(T) = FIRST(F) \\ &= \{ (, id \} \end{aligned}$$

$$FIRST(E') = \{ +, \epsilon \}$$

$$FIRST(T') = \{ *, \epsilon \}$$

→ Compute follow

(i) If A is a start symbol, put $\$$ in $FOLLOW(A)$

(ii) Production of the form $B \rightarrow \alpha A \beta$

$$FOLLOW(A) = FIRST(\beta)$$

(iii) Production of the form $B \rightarrow \alpha A$ or $B \rightarrow \alpha A \beta$, where $\beta \xrightarrow{*} \epsilon$, add

$$FOLLOW(A) = FOLLOW(B)$$

$$FOLLOW(E) = \{ \$,) \}$$

$$\begin{aligned} FOLLOW(E') &= FOLLOW(E') + FOLLOW(E) \\ &= \{ \$,) \} \end{aligned}$$

$$\begin{aligned} FOLLOW(T) &= FIRST(E') \\ &= \{ +, \epsilon \} \end{aligned}$$

$$\begin{aligned} &\hookrightarrow FOLLOW(E) \\ &= \{ +, \$,) \} \end{aligned}$$

$$\begin{aligned} FOLLOW(T') &= FOLLOW(T') + FOLLOW(T) \\ &= \{ +, \$,) \} \end{aligned}$$

$$FOLLOW(F) = \{ *, +, \$,) \}$$

$$\alpha A \beta$$
$$F \rightarrow (E)$$

$$FOLLOW(E) = FIRST(\beta)$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE'$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE'$$

$$③ S \rightarrow (A) / \epsilon$$

$$A \rightarrow TE$$

$$E \rightarrow \&TE / \epsilon$$

$$T \rightarrow (A) / a / b / c$$

$$\text{FIRST}(S) = \{ (, \epsilon \}$$

$$S \rightarrow \underline{(A)} / \underline{\epsilon}$$

$$\text{FIRST}(A) = \{ (, a, b, c \}$$

$$\left\{ \begin{array}{l} A \rightarrow TE \\ T \rightarrow \underline{(A)} / \underline{a} / \underline{b} / \underline{c} \end{array} \right\}$$

$$\text{FIRST}(E) = \{ \&, \epsilon \}$$

$$E \rightarrow \underline{\&}TE / \underline{\epsilon}$$

$$\text{FIRST}(T) = \{ (, a, b, c \}$$

$$T \rightarrow \underline{(A)} / \underline{a} / \underline{b} / \underline{c}$$

$$\text{FOLLOW}(S) = \{ \$ \}$$

$$\text{FOLLOW}(A) = \{) \}$$

$$\text{FOLLOW}(E) = \text{FOLLOW}(A) = \{) \}$$

$$A \rightarrow TE$$

$$E \rightarrow \&TE / \epsilon$$

$$\text{FOLLOW}(T) = \text{FIRST}(E)$$

$$A \rightarrow T\epsilon$$

$$= \{ \&, \epsilon \}$$

$$E \rightarrow \&T\epsilon$$

$$\hookrightarrow \text{FOLLOW}(E)$$

$$= \{ \&,) \}$$

$$4. S \rightarrow ABCDE$$

$$A \rightarrow a/\epsilon$$

$$B \rightarrow b/\epsilon$$

$$C \rightarrow c$$

$$D \rightarrow d/\epsilon$$

$$E \rightarrow e/\epsilon$$

$$\text{FIRST}(A) = \{a, \epsilon\}$$

$$\text{FIRST}(B) = \{b, \epsilon\}$$

$$\text{FIRST}(C) = \{c\}$$

$$\text{FIRST}(D) = \{d, \epsilon\}$$

$$\text{FIRST}(E) = \{e, \epsilon\}$$

$$\text{FIRST}(S) = \text{FIRST}(ABCDE)$$

$$= \{a, \epsilon\}$$

$$\hookrightarrow \text{FIRST}(BCD)$$

$$= \{a, b, \epsilon\}$$

$$\hookrightarrow \text{FIRST}(CD)$$

$$= \{a, b, c, \epsilon\}$$

$$\hookrightarrow \text{FIRST}(D)$$

$$= \{a, b, c, d, \epsilon\}$$

$$= \{a, b, c, d, e\}$$

$$\text{FOLLOW}(S) = \{\$ \}$$

$$A \xrightarrow{\epsilon} BCDE$$

$$\text{FOLLOW}(A) = \text{FIRST}(B)$$

$$= \{b, \epsilon\}$$

$$\hookrightarrow \text{FIRST}(C)$$

$$= \{b, c\}$$

$$\text{FOLLOW}(B) = \text{FIRST}(C) = \{c\}$$

$$\text{FOLLOW}(C) = \text{FIRST}(D) = \{d, \epsilon\} = \{d, e, \$ \}$$

$$\hookrightarrow \text{FIRST}(E)$$

$$\text{FOLLOW}(S)$$

$$\text{FOLLOW}(D) = \text{FIRST}(E) = \{e, \epsilon\} = \{e, \$ \}$$

$$\text{FOLLOW}(E) = \text{FOLLOW}(S) = \{\$ \}$$

5. $A \rightarrow aBa$
 $B \rightarrow bB/\epsilon$

$$\text{FOLLOW}(A) = \{\$ \}$$

$$\text{FOLLOW}(B) = \{a\}$$

$$\text{FIRST}(A) = \{a\}$$

$$\text{FIRST}(B) = \{b, \epsilon\}$$

6. $S \rightarrow Bb|Cd$

$$B \rightarrow aB/\epsilon$$

$$C \rightarrow cC/\epsilon$$

$$\text{FIRST}(B) = \{a, \epsilon\}$$

$$\text{FIRST}(C) = \{c, \epsilon\}$$

$$\begin{aligned} \text{FIRST}(S) &= \text{FIRST}(Bb) + \text{FIRST}(Cd) \\ &= \{a, b, c, d\} \end{aligned}$$

$$\text{FOLLOW}(S) = \{\$ \}$$

$$\text{FOLLOW}(B) = \{b\}$$

$$\text{FOLLOW}(C) = \{d\}$$

8. Calculate first and follow functions for the given grammar.

$$S \rightarrow aBDh$$

$$B \rightarrow cC$$

$$C \rightarrow bC/\epsilon$$

$$D \rightarrow EF$$

$$E \rightarrow g/\epsilon$$

$$F \rightarrow f/\epsilon$$

$$\text{FIRST}(S) = \{a\}$$

$$\text{FIRST}(B) = \{c\}$$

$$\text{FIRST}(C) = \{b, \epsilon\}$$

$$\begin{aligned}\text{FIRST}(D) &= \{(\text{FIRST}(E) - \epsilon) \cup \text{FIRST}(F)\} \\ &= \{g, f, \epsilon\}\end{aligned}$$

$$\text{FIRST}(E) = \{g, \epsilon\}$$

$$\text{FIRST}(F) = \{f, \epsilon\}$$

$$\text{FOLLOW}(S) = \{\$ \}$$

$$\begin{aligned}\text{FOLLOW}(B) &= \{\text{FIRST}(D) - \epsilon\} \cup \text{FIRST}(h) \\ &= \{g, f, h\}\end{aligned}$$

$$\begin{aligned}\text{FOLLOW}(C) &= \text{FOLLOW}(B) \\ &= \{g, f, h\}\end{aligned}$$

$$\text{FOLLOW}(D) = \{h\}$$

$$\text{FOLLOW}(E) = \{\text{FIRST}(F) - \epsilon\} \cup \text{FOLLOW}(D) = \{f, h\}$$

$$\begin{aligned}\text{FOLLOW}(F) &= \text{FOLLOW}(D) \\ &= \{h\}\end{aligned}$$